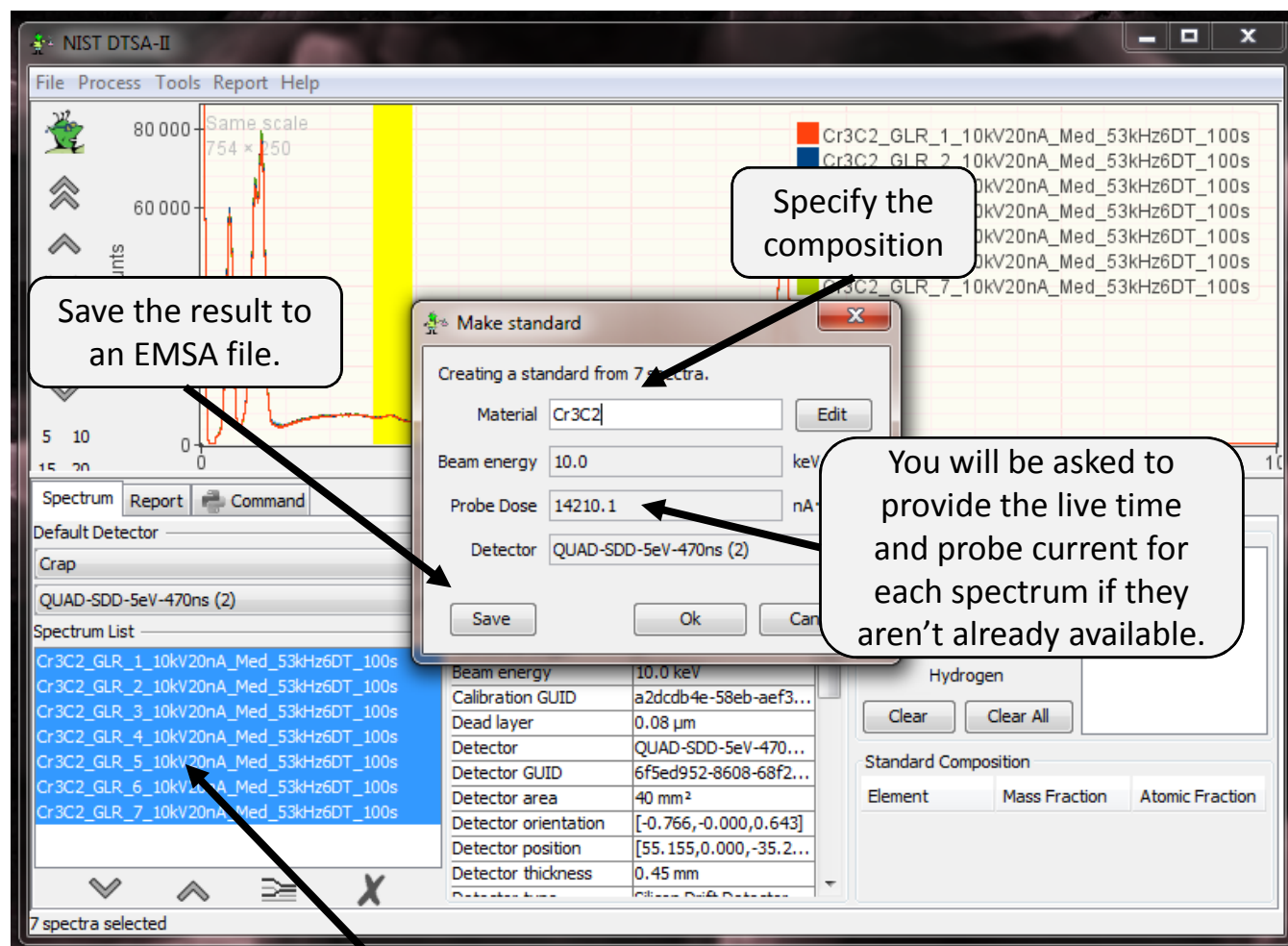


DTSA-II: Utilities

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Make standard

Collecting many shorter spectra from a standard is usually a better strategy than collecting one longer spectrum because you can compare the shorter spectra and discard ones that have anomalies.

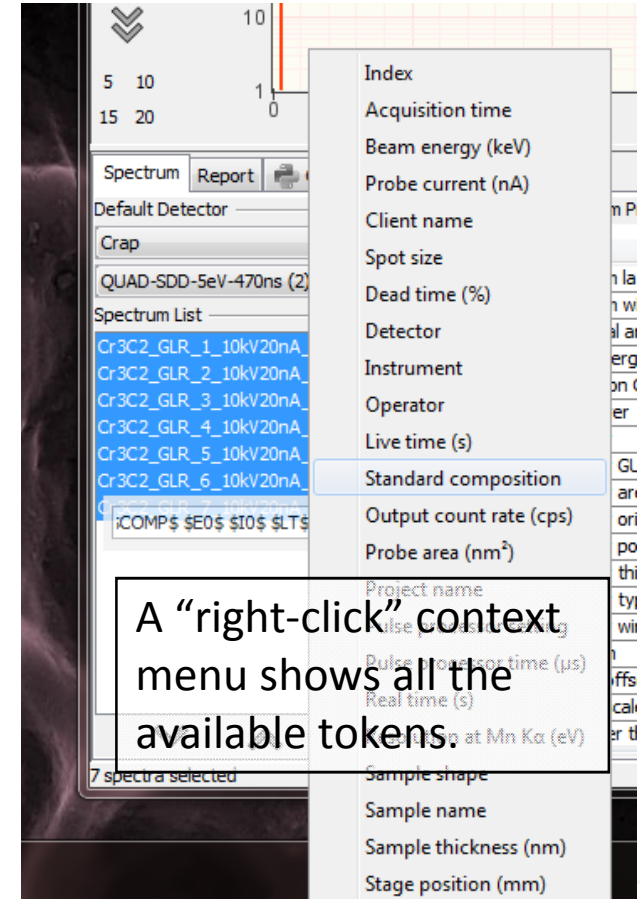
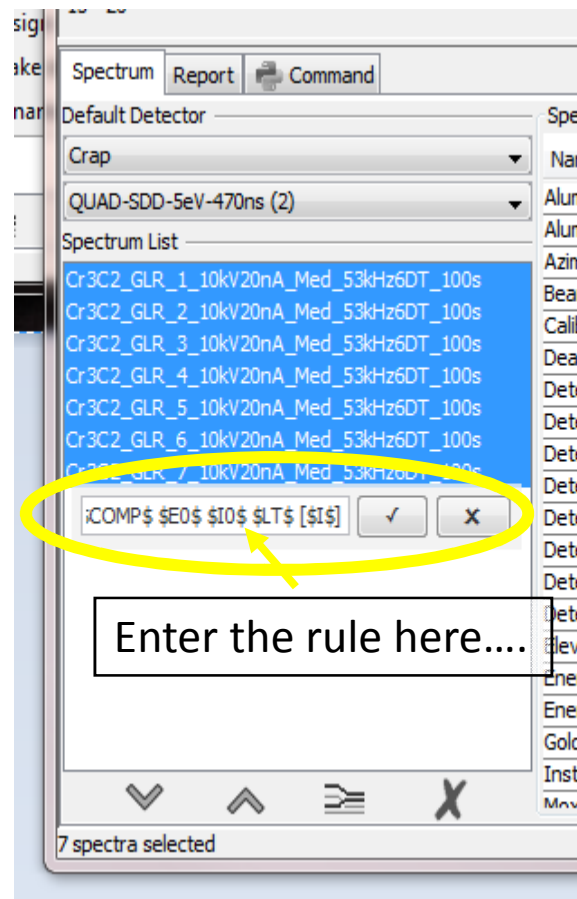
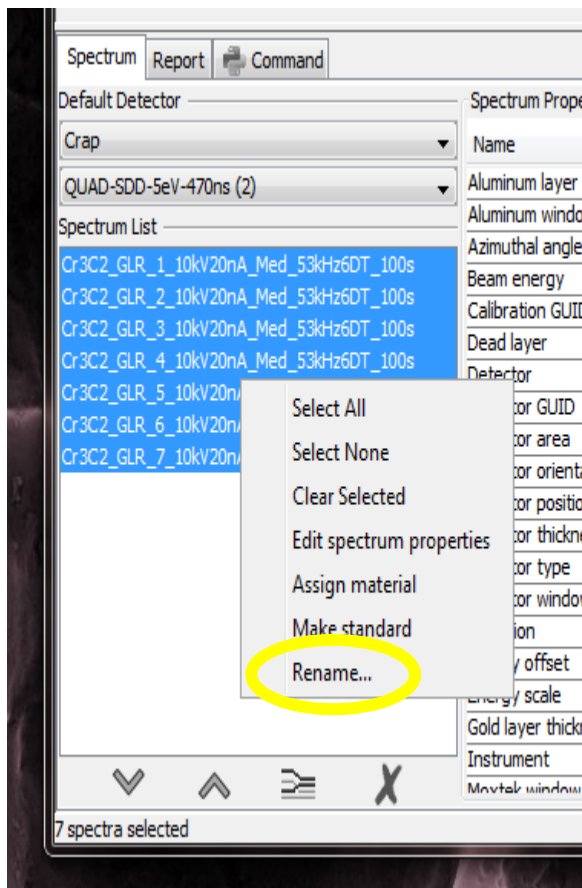


This is a utility to assist turning raw spectra into standards. It will combine all the selected spectra into a single standard by summing together the counts. The utility also ensures that all necessary pieces of information are included in the standard spectrum.

“Make standard” is available from the Spectrum List context (“right click”) menu.

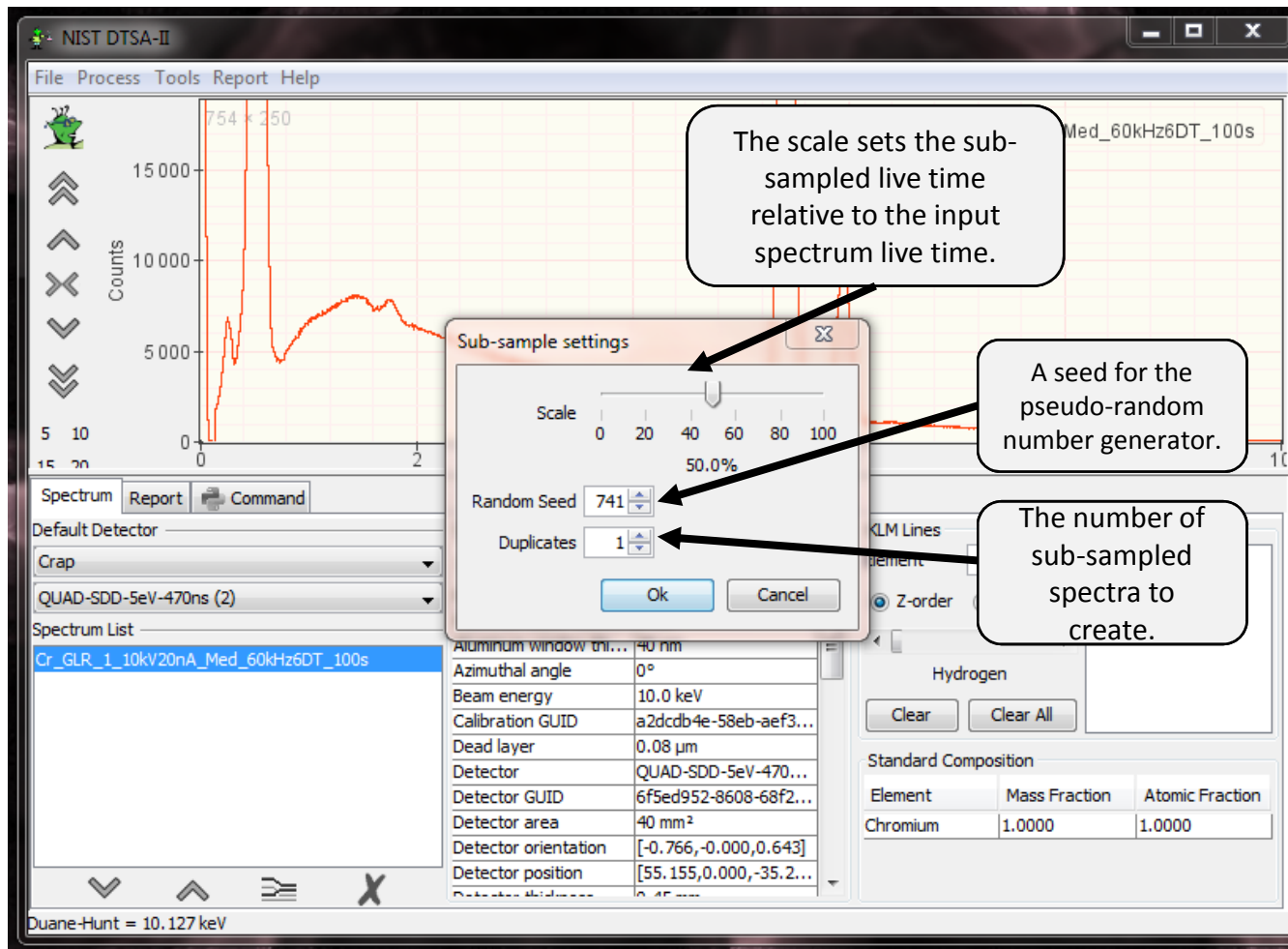
Renaming spectra

It is simple and quick to rename spectra using the spectrum rename utility. The utility takes a user defined rule consisting of verbatim text and replaceable tokens. The tokens allow you to rename spectra based on spectrum properties.



Menu item: Process -> Sub-sample spectrum

Sub-sampling spectra

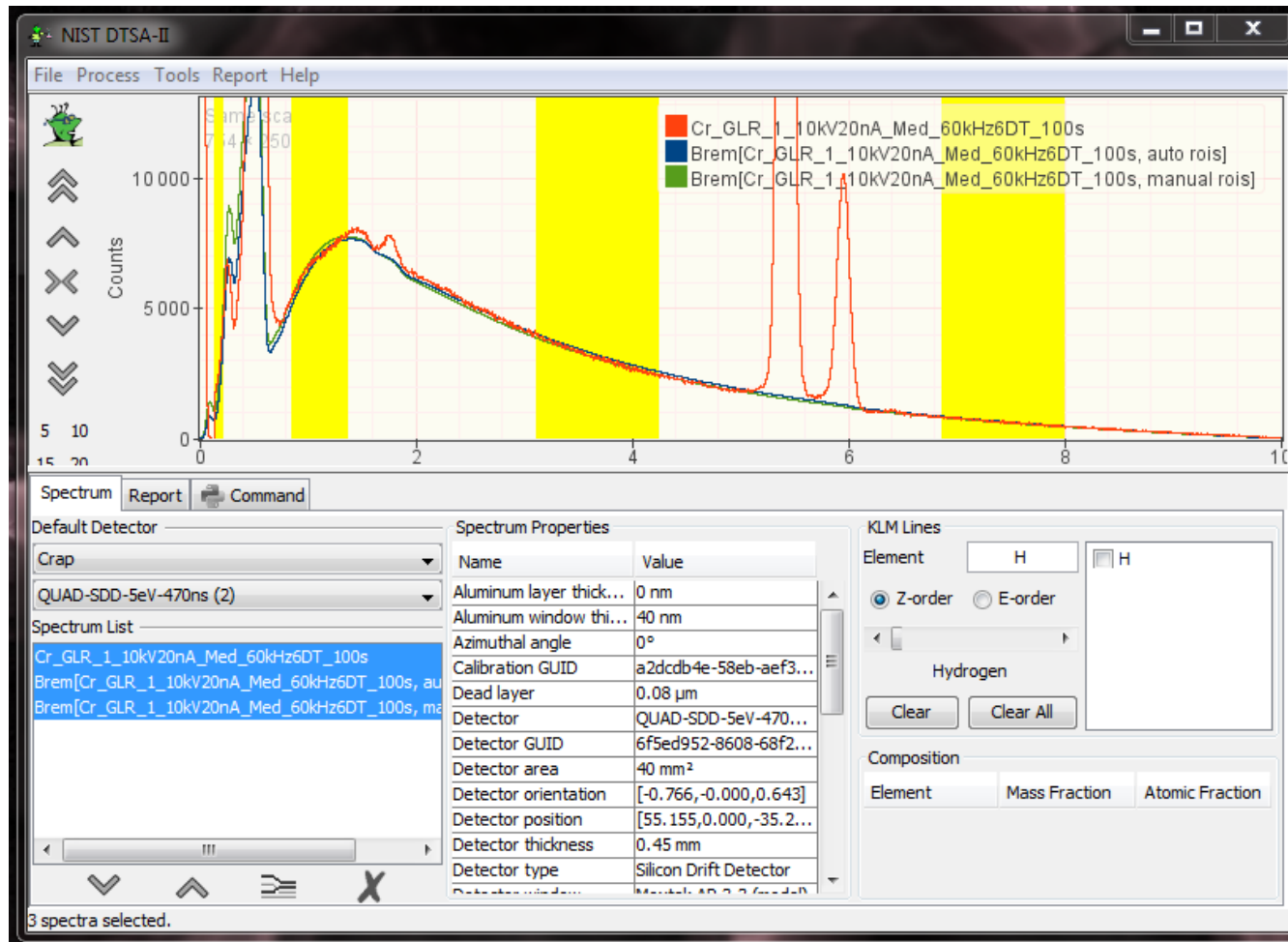


Sometimes it is interesting to know how reducing the number of counts in a spectrum will effect the precision of measurements.

This function does a statistically correct sub-sampling of a measured spectrum. The result is as though the spectrum was collected for a shorter live time.

Menu item: Process -> Fit background

Fit background



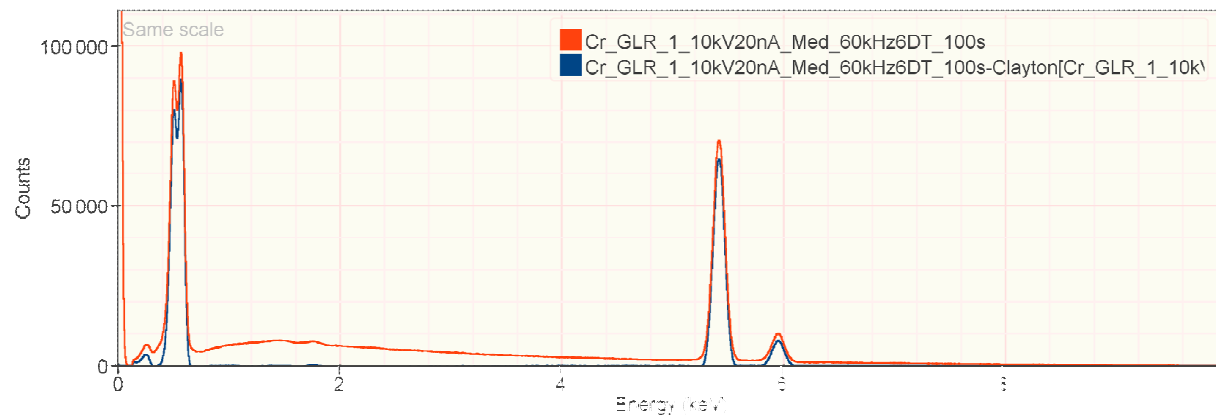
Fit a modeled Bremsstrahlung / continuum to the selected spectra. The model needs to know the composition of the material which can be specified via quantification or via assigning a standard composition.

Modes:

1. Auto-ROIs – Picks regions for the fit based on the material composition
2. Manual ROIs – Specify regions on the spectrum display window (yellow ranges)

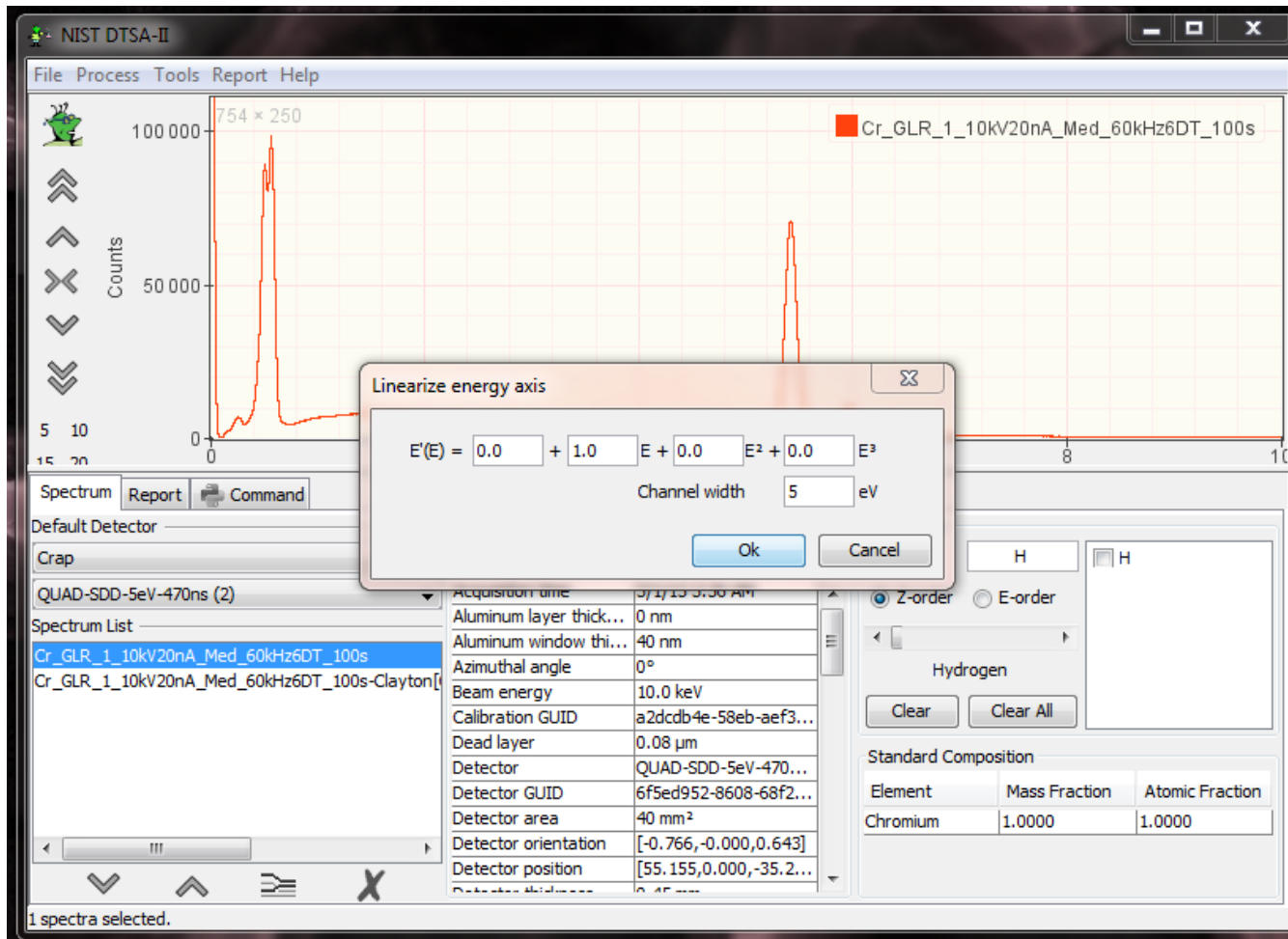
Strip background

Uses a non-physics-based algorithm (Clayton, 1987) to strip the Bremsstrahlung / continuum from a spectrum. Use “Fit background” and then spectrum math to perform a physics-based background elimination.



Menu item: Process -> Linearize Energy Axis

Linearize energy axis

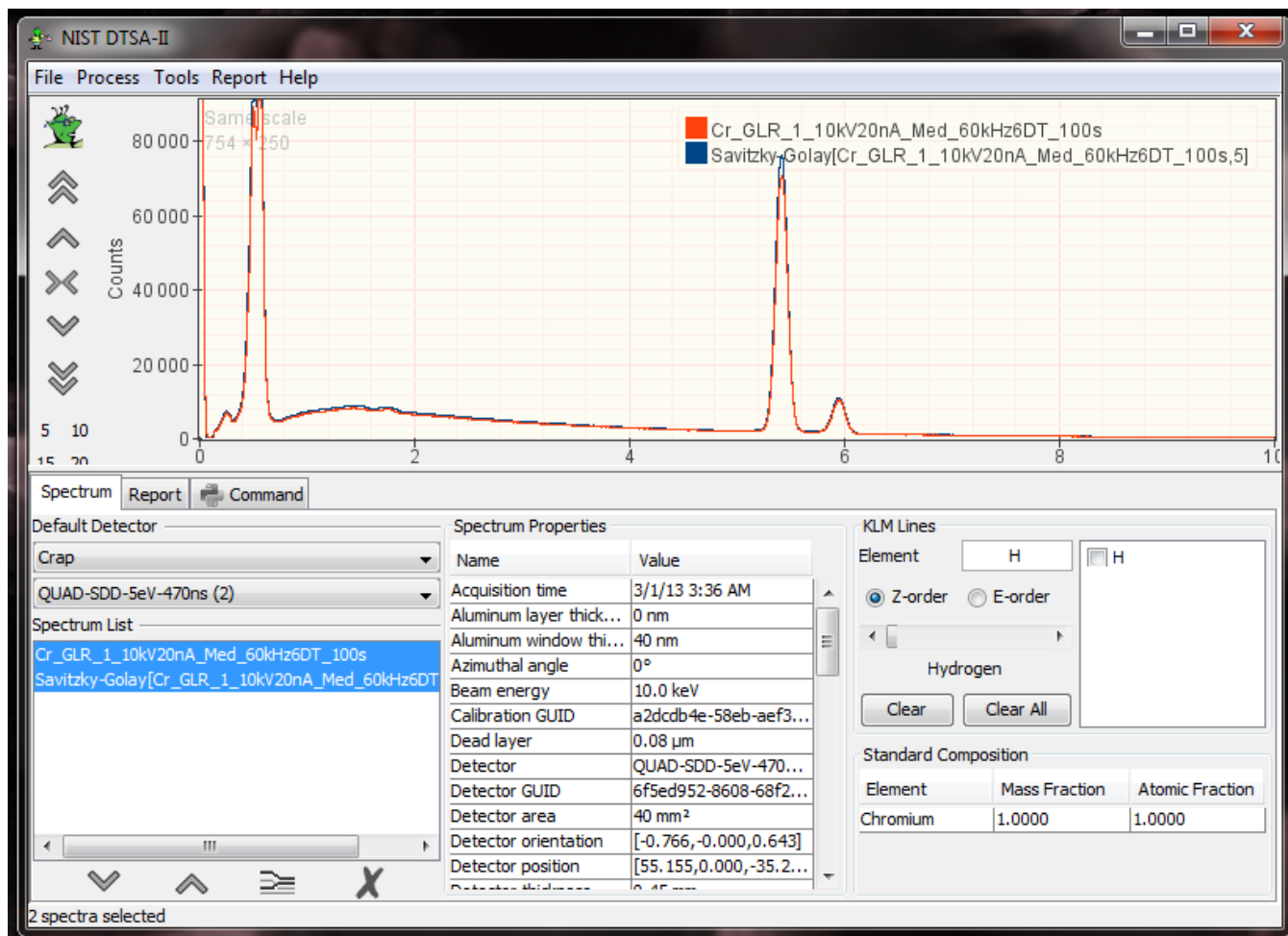


This is a very specialized tool for correcting non-linearity in the energy axis up to a cubic polynomial energy scale. It can also be used to decrease the number of bins (increase the channel width.)

Menu item: Process -> Smooth (Savitsky-Golay)

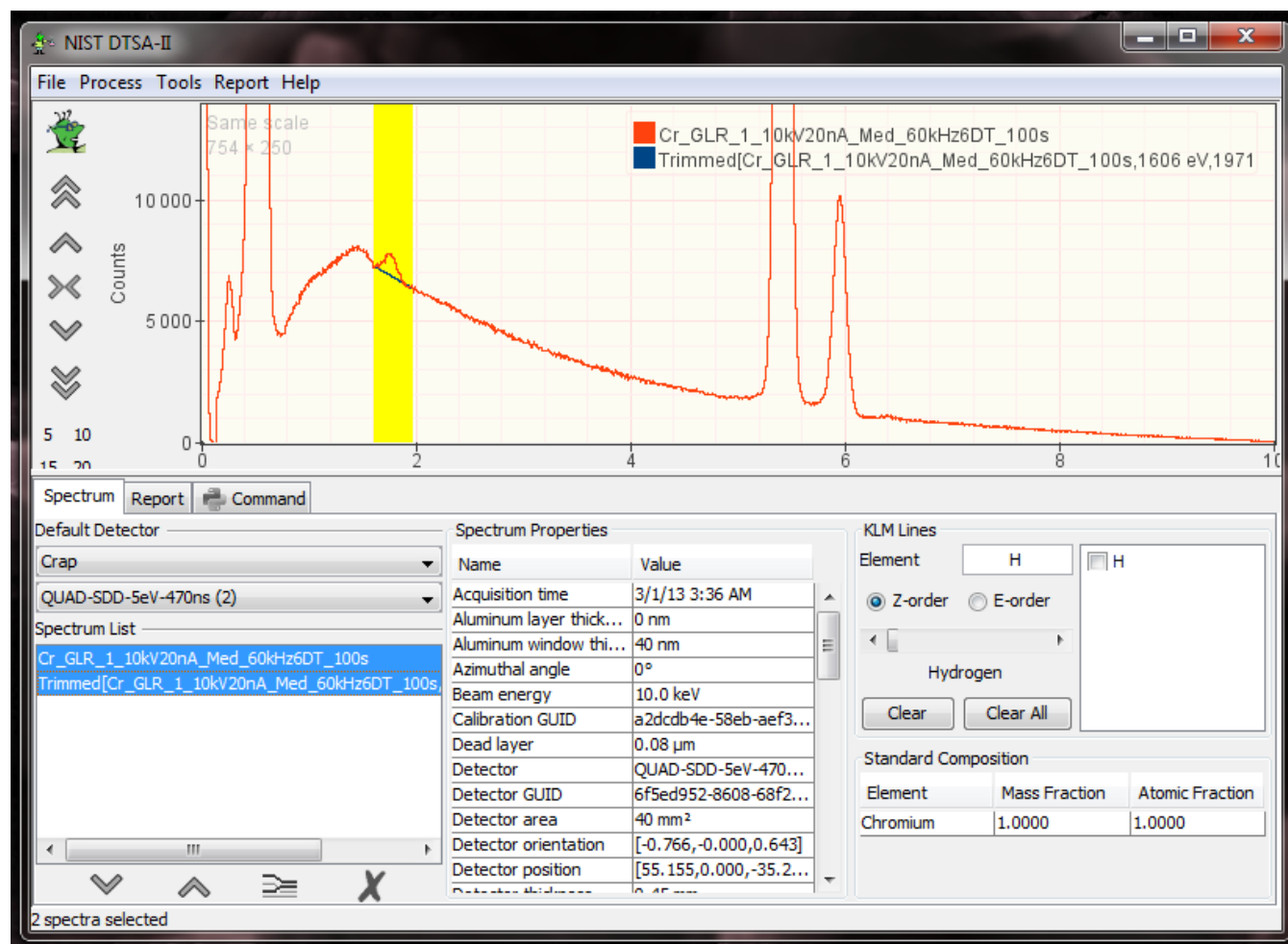
Smooth (Savitsky-Golay)

Use judiciously. If you find you need to smooth a spectrum for presentation purposes, there is probably a problem. You probably should have collected for a longer live time.



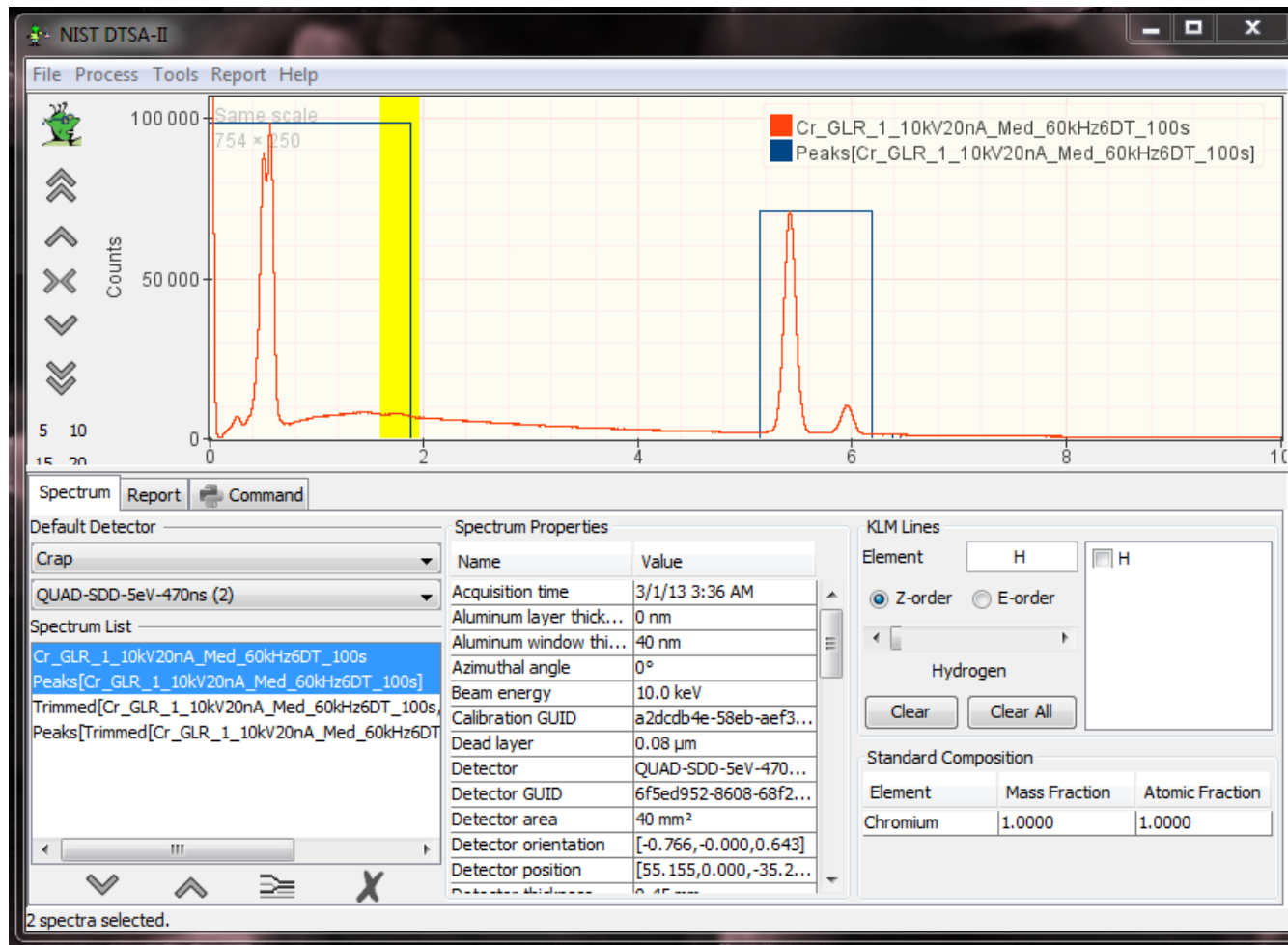
Menu item: Process -> Trim

Trim



Sometimes you may have a spectrum that is almost a good reference except for the presence of an unfortunately located minor peak. This function can be used to eliminate the peak by fitting a line between the two edges of the selected region(s).

Peak search



Identifies regions of the spectrum which potentially could have statistically significant peaks.